

Requirement Analysis - DFD & User Stories

Parameter	Detail
Date	18 June 2026
Team ID	Book-a-Doctor Platform
Project Name	Book a Doctor
Maximum Marks	4 Marks

Data Flow Diagrams

DFD Level 0 (Context Diagram)



DFD Level 1 (Process Breakdown Diagram)

```
graph TD
    User([User]) -->|Auth Request| P1[1.0 Authentication Engine]
    P1 -->|Query/Save| DB_User[(MongoDB Users Collection)]

    Patient([Patient]) -->|Booking Request| P2[2.0 Scheduling Engine]
    P2 -->|Validation check| DB_Appt[(MongoDB Appointments Collection)]

    Patient -->|Upload File| P3[3.0 Medical Locker Service]
    P3 -->|Save file metadata| DB_Report[(MongoDB Reports Collection)]
    P3 -->|Save file to disk| LocalStorage[Local uploads/ folder]

    Doctor([Doctor]) -->|Submit Prescription| P4[4.0 Prescription Service]
    P4 -->|Write Prescription| DB_Appt

    Admin([Admin]) -->|Approve Doctor| P5[5.0 Admin Vetting Engine]
    P5 -->|Update Doctor status| DB_User
```

User Stories

User Type	Epic / Feature	Story ID	User Story / Task	Acceptance Criteria	Priority	Release
Patient	Authentication	USN-01	As a patient, I want to sign up and log in securely so that I can access my booking dashboard.	- Bcrypt password hashing - Valid JWT access token returned	High	Sprint-1
Doctor	Registration	USN-02	As a doctor, I want to register with details (fee, specialty) so that the admin can vet my credentials.	- Saved in "pending" status - Not visible in Patient search	High	Sprint-1
Patient	Search Engine	USN-03	As a patient, I want to filter doctors by specialty and fee so that I can find a match.	- Mongoose regex filters - Shows only "approved" doctors	High	Sprint-2
Patient	Booking Engine	USN-04	As a patient, I want to book a consultation slot so that I can secure doctor availability.	- Slot double-booking prevention - Saved with "pending" status	High	Sprint-2

User Type	Epic / Feature	Story ID	User Story / Task	Acceptance Criteria	Priority	Release
Admin	Admin Panel	USN-05	As an admin, I want to approve doctor profiles in the vetting queue so that they can take bookings.	- Admin dashboard vetting tab - Toggles doctor approval status	High	Sprint-3
Doctor	Doctor Panel	USN-06	As a doctor, I want to approve/reject slot requests so that I can schedule my workload.	- Action column on dashboard - Notification log updated	High	Sprint-3
Patient	Locker	USN-07	As a patient, I want to upload my health records so that my doctor can inspect them.	- Only PDF/JPG files accepted - Stored in uploads/directory	Medium	Sprint-3
Doctor	Prescriptions	USN-08	As a doctor, I want to write prescriptions for approved slots so that the patient receives them.	- Prescription save modal - Changes appointment to "completed"	High	Sprint-4
Patient	Notifications	USN-09	As a user, I want to see notification alerts when my status changes so that I stay informed.	- Notification collection query - Unread toggle on frontend	Medium	Sprint-4